ECON 3201
Econometrics for Economics and Finance

Assignment 1
Due Thursday October 8, 2026

Your_Name_Here

Instructions

Answer all questions. Please complete your assignment using Quarto. Submit both your Quarto file and pdf output file.

Theory Questions

Q1

Create a personal Quarto website hosted on GitHup pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

i. An "Index" or "Home" page. This page should include the following:

- Your name
- Your photo (optional)
- Your credentials
- An "About Me" section that provides a brief (one paragraph) introduction to yourself. [10 Marks]

ii. A blog page. This will be the landing site for future assignments [5 Marks]
iii. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
iv. Navigation bar with "Home" and "Blog" buttons. The navigation bar is edited in the "_quarto.yml" file. [2 Marks]

> Paste your link here

Q2

Write your solutions using LaTeX. Show your work. You may use R to compute the answers.

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Check that $f_X(x)$ is a true probability distribution.

Paste your answer here

(b) Compute the expected value of X.

Paste your answer here

(c) Compute the variance of X.

Paste your answer here

Q3

Let $X \sim N(4, 5)$. Find the probability of each event:

(a) $Pr(X \leq 6)$

Paste your answer here

(b) $Pr(X > 4)$

Paste your answer here

(c) $Pr(-1 \leq X < 1)$

Paste your answer here

Q4

Let $Z \sim N(0, 1)$.

(a) Compute $Pr(Z \leq 0.83)$

Paste your answer here

(b) Compute $Pr(-0.83 \leq Z \leq 0.83)$

Paste your answer here

R Excercises

Q5

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Plot the probability distribution function $f_X(x)$ in R.

Paste your answer here

(b) Plot the cumulative probability distribution function $f_X(x) in R$.

Paste your answer here

(c) Compute the expected value of X in R.

Paste your answer here

(d) Compute the variance of X in R.

Paste your answer here

Q6

The standard normal distribution function is given by

$$f(x) = \frac{e^{-(x^2/2)}}{\sqrt{2\pi}\sigma}$$

(a) Define this function in R. (Note: π is coded as `pi` and e is coded as `exp`.

Paste your answer here

(b) Integrate this function over the limit $[-\infty, \infty]$ using the `integrate()` command in R. Infinity in R is written as `Inf`. To evaluate the integral, use `integrate()$value`

Paste your answer here

(c) Compute the expected value of the standard normal distribution in R and confirm it is equal to 0.

Paste your answer here

(d) Compute the variance of the standard normal distribution in R and confirm it is equal to 1.

Paste your answer here

(e) Compute $Pr(Z \leq 0.83)$

Paste your answer here

(f) Compute $Pr(-0.83 \leq Z \leq 0.83)$

Paste your answer here

(g) Compute $Pr(-0.83 \leq Z \leq 0.83)$ using the `pnorm()` command and compare your results to (f).

Paste your answer here

Q7

Let X denote the prison sentence, in years, for people convicted of auto theft in a particular province in Canada. Suppose the PDF of X is given by:

$$f(X) = \frac{1}{9}x^2, \ 0 < x < 3.$$

(a) Define $f(x)$ in R.

Paste your answer here

(b) Integrate $f(x)$ over the interval [0,3].

Paste your answer here

(c) Compute the expected prison sentence.

Paste your answer here

Q8

Let X follow a Student t distribution with M degrees of freedom. Then

$$X = \frac{Z}{\sqrt{W/M}} \sim t_M,$$

where Z is a standard normal variable and W is a chi-squared variable with M degrees of freedom.

(a) On the same figure plot the following:

(i) Plot the standard normal probability distribution using `curve()`. Specify the range as [-5,5] using `xlim`. Name the `ylab` "Density". The main title should be "Densities of t Distribution".

(ii) Plot the Student t distribution with 5 degrees of freedom using `curve()` and `dt()`. Specify the range as [-5,5] using `xlim`. Choose a different colour, i.e. `col=2`. To learn more about `dt`, type `?dt` in your console.

(iii) Plot the Student t distribution with 15 degrees of freedom using `curve()` and `dt()`. Specify the range as [-5,5] using `xlim`. Choose a different colour, i.e. `col=3`. To learn more about `dt`, type `?dt` in your console.

(iv) Plot the Student t distribution with 25 degrees of freedom using `curve()` and `dt()`. Specify the range as [-5,5] using `xlim`. Choose a different colour, i.e. `col=4`. To learn more about `dt`, type `?dt` in your console.

(v) Include the following legend in you plot:
`legend("topright", c("N(0, 1)", "M=2", "M=4", "M=25"), col = 1:4, lty = c(1, 1, 1, 1))`

Paste your answer here

(b) What do you observe as the degrees of freedom increase?

Paste your answer here